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GUT CONTENTS OF JUVENILE CHINOOK SALMON FROM THE UPPER SACRAMENTO RIVER, CALIFORNIA, DURING SPRING 1998

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During the planning phase of a field investigation on heavy-metal contamination of chinook salmon, *Oncorhynchus tshawytscha*, and their food supply in the upper Sacramento River (see Saiki¹ 2000 for findings from that investigation), we found only 4 references dealing with food habits of juvenile salmon in the river and its tributaries below Keswick Dam. Rutter (1904) recorded the gut contents of 210 fish from the Sacramento River and its tributaries between Sullaway Creek (located upstream from Dunsmuir) and Rio Vista during 1898–1899. Schaffter et al.² (1983) summarized the gut contents of 466 juvenile salmon from the Sacramento River between Red Bluff and Chico during 1981. Merz and Vanicek (1996) described the gut contents of 468 juvenile salmon from the lower American River between Nimbus Dam and its confluence with the Sacramento River during 1992–1993. Finally, Moore³ (1997) examined the gut contents of 16 juvenile salmon from the Sacramento River at Red Bluff and 150 juvenile salmon from Dibble and Blue Tent creeks, 2 intermittent tributaries that enter the Sacramento River immediately upstream from Red Bluff. Although Ganssle (1966), Sasaki (1966), Kjelson et al. (1982), and others documented the gut contents of juvenile salmon from the Sacramento-San Joaquin Delta and San Francisco Bay, the forage base in this estuarine habitat can differ considerably from that upstream in the Sacramento River. The purpose of our study was to supplement existing information on forage taxa consumed by juvenile salmon in the upper Sacramento River by documenting their gut contents.

In March and May 1998, sampling sites were established along the upper Sacramento River as follows (from upstream to downstream) (Fig. 1): adjacent to Jellys Ferry Road (Jellys Ferry; N40°20'54.1", W122°10'55.0"); at Red Bluff Lake

¹ Saiki, M.K. 2000. Assessment of copper, cadmium, and zinc contamination in juvenile chinook salmon and selected fish-forage organisms (aquatic insects) in the upper Sacramento River, California. Final report prepared for U.S. Fish and Wildlife Service, Sacramento Fish and Wildlife Office, Sacramento, California, USA.

² Schaffter, R.G., P.A. Jones, and J.G. Karlton. 1983. Sacramento River and tributaries bank protection and erosion control investigation – evaluation of impacts on fisheries: Final report. California Department of Fish and Game, Bay-Delta Fishery Project, Stockton, California, USA.

³ Moore, T.L. 1997. Condition and feeding of juvenile chinook salmon in selected intermittent tributaries of the upper Sacramento River. MS Thesis, California State University, Chico, California, USA.

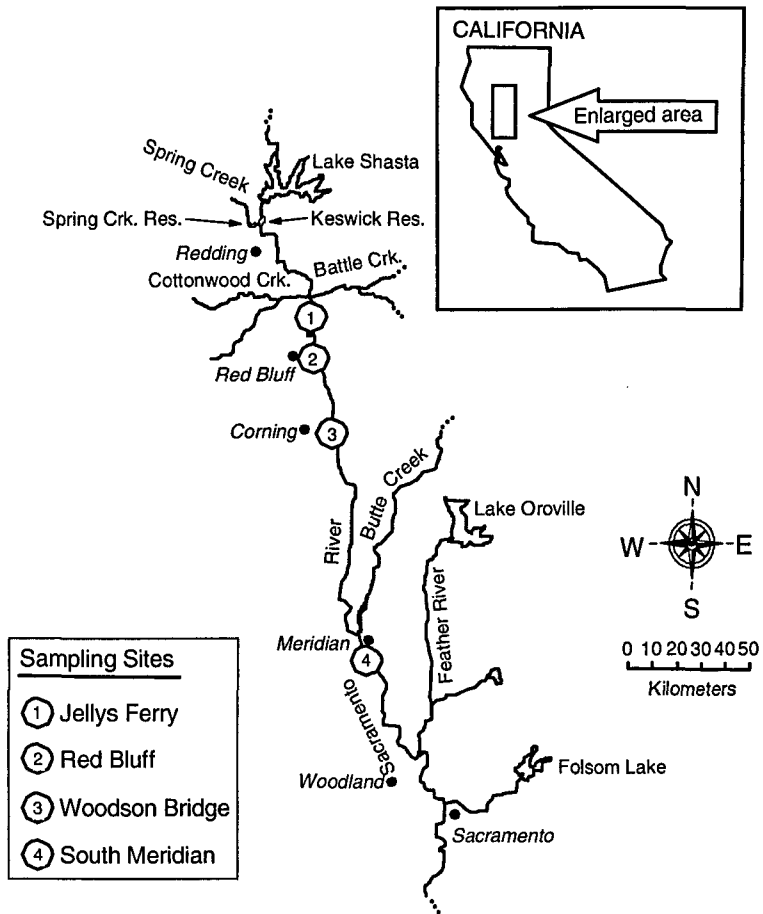


Figure 1. Map of the study area with locations of sampling sites on the upper Sacramento River.

(Red Bluff; $N40^{\circ}09'19.0''$, $W122^{\circ}12'23.6''$); near the City of Corning (Woodson Bridge; $N39^{\circ}54'31.9''$, $W122^{\circ}05'29.0''$); and below the City of Meridian (South Meridian; $N39^{\circ}06'47.2''$, $W121^{\circ}54'12.2''$). Juvenile chinook salmon were captured from these sites with beach seines (3.2-mm bar mesh). Fish were measured to the nearest millimeter standard length (SL), weighed to the nearest 0.01 g, coded with unique identification numbers, and preserved in 10% formalin. About 4 days later, the fish were transferred into 70% isopropyl alcohol and stored until gut analysis could be performed. Although fin-clipped chinook salmon were excluded from our study, some fish may have originated from the Coleman National Fish Hatchery because, of approximately 12 million salmon released into Battle Creek during March–April 1998, only 1 million fish were fin clipped. However, studies elsewhere have found that hatchery salmonids adapt to wild prey as rapidly as 1 week after release (Johnsen and Ugedal 1986, Reirz et al. 1998), so potential inclusion of some hatchery chinook salmon in our sample should not bias results.

Stomach contents from the anterior end of the esophagus to the pyloric sphincter were identified to the lowest taxonomic level possible using standard keys (Unger 1971, Merritt and Cummins 1978, Pennak 1978). Damp-dry biomass (nearest 0.00001 g) of each taxonomic category was recorded for individual fish. Percent damp-dry biomass of each taxon consumed by fish from a given site was calculated as 100 times the sum of the biomass of that taxon for all fish from the site divided by the sum of the biomass of all taxa for all fish from the site.

A total of 114 juvenile chinook salmon averaging 56 ± 9 mm SL (weight, 2.78 ± 1.11 g; values are arithmetic mean $\pm$ SD) was collected during this study: 25 came from Jellys Ferry (all in May), 26 from Red Bluff (all in March), 36 from Woodson Bridge (11 in March, 25 in May), and 27 from South Meridian (2 in March, 25 in May). Judging from daily fork length criteria for 4 runs of juvenile salmon in the Sacramento River (Fisher⁴ 1992, as modified by Greene⁵ 1992), fish used in our study were most likely progeny of fall-run adults.

Gut contents of juvenile chinook salmon captured in March and May were combined after preliminary comparisons of mean damp-dry biomasses of the forage taxa ingested by fish from Woodson Bridge and South Meridian yielded overlapping 95% confidence intervals. Only 2 fish from Red Bluff, 1 fish from Woodson Bridge, and 1 fish from South Meridian had empty guts.

On average, the most important forage of juvenile chinook salmon were midge (Diptera: Chironomidae) larvae and pupae, caddisfly (Trichoptera: Hydropsychidae) larvae, and amorphous organic matter (mostly unidentifiable insect remains) (Table 1). At Jellys Ferry, gut contents consisted mostly of midge larvae and pupae (77.2%) and Cladocera (12.1%), whereas at Red Bluff, amorphous organic matter (25.9%), hydropsychid caddisfly larvae (21.1%), and Hemiptera (11.4%) were the most important forage taxa. At both Woodson Bridge and South Meridian, juvenile salmon consumed mostly hydropsychid caddisfly larvae (Woodson Bridge, 25.5%; South Meridian, 34.5%) and amorphous organic matter (Woodson Bridge, 22.9%; South Meridian, 16.5%). Midge larvae and pupae (15.9%) were also important at Woodson Bridge, whereas mayfly (Ephemeroptera: Baetidae) nymphs (13.5%) were important at South Meridian. Other taxa contributed <10% of overall damp-dry biomass of gut contents.

Although many studies are available on the food habits of juvenile chinook salmon in fresh waters of the Pacific Northwest, including northern California and Alaska (e.g., Loftus and Lenon 1977, Busby and Barnhart 1995, Muir and Coley 1996), and elsewhere in the world (e.g., Johnson 1981, Field-Dodgson 1988, Sagar and Glova 1988, Rutledge and Power 1992, Smirnov et al. 1994), few studies have occurred in the upper Sacramento River drainage. According to an early report by Rutter (1904),

⁴Fisher, F.W. 1992. Chinook salmon *Oncorhynchus tshawytscha* growth and occurrence in the Sacramento-San Joaquin River system. California Department of Fish and Game, Inland Fisheries Division, Red Bluff, California, USA.

⁵Greene, S. 1992. Estimated winter-run chinook salmon salvage at the State Water Project and Central Valley Project delta pumping facilities. Memorandum to R. Brown, 5/8/92. California Department of Water Resources, Sacramento, California, USA.

Table 1. Damp-dry biomass of gut contents from 114 juvenile chinook salmon at 4 sites on the upper Sacramento River, spring 1998.

Taxonomic category	Jellys Ferry (n = 25)		Red Bluff (n = 26)		Woodson Bridge n = 36)		South Meridian (n = 27)	
	Grams	%	Grams	%	Grams	%	Grams	%
Coleoptera:								
Larvae	0.00040	0.09	0.01770	6.16	0.00670	2.39	0.01310	4.44
Adult	0.00080	0.18	0.02590	9.01	0.02530	9.02	0.00320	1.08
Diptera:								
Chironomidae	0.34791	77.20	0.00890	3.10	0.04460	15.90	0.02522	8.55
Larvae and pupae								
Other Diptera	0.00350	0.78	0.00390	1.36	0.01560	5.56	0.01860	6.30
Larvae and pupae								
Diptera adults	0.00130	0.29	0.01310	4.56	0.00290	1.03	0.00060	0.20
Ephemeroptera:								
Baetidae nymphs	0.00330	0.73	0.00490	1.71	0.01370	4.88	0.03970	13.45
Other mayfly nymphs	0.00001	<0.01	0.00000	0.00	0.00000	0.00	0.00000	0.00
Mayfly adults	0.00000	0.00	0.00000	0.00	0.00210	0.75	0.00000	0.00
Hemiptera	0.00090	0.20	0.03260	11.35	0.01230	4.38	0.01410	4.78
Trichoptera:								
Hydropsychidae larvae	0.01580	3.51	0.06050	21.06	0.07160	25.53	0.10180	34.50
Other caddisfly larvae	0.00082	0.18	0.00220	0.76	0.00040	0.14	0.00000	0.00
Caddisfly adults	0.00000	0.00	0.00000	0.00	0.00000	0.00	0.00370	1.25
Other insects	0.01700	3.77	0.02130	7.41	0.01520	5.42	0.01240	4.20
Cladocera	0.05460	12.12	0.00000	0.00	0.00000	0.00	0.00000	0.00
Other invertebrates	0.00060	0.13	0.01200	4.18	0.00590	2.10	0.01270	4.31
Amorphous matter	0.00370	0.82	0.07440	25.90	0.06420	22.89	0.04860	16.47
Sand	0.00000	0.00	0.00990	3.45	0.00000	0.00	0.00140	0.47
Total	0.45064	100.00	0.28730	100.00	0.28050	100.00	0.29512	100.00

juvenile chinook salmon in the Sacramento River and its tributaries between Redding and Grimes (the approximate longitudinal extent of our study area) consumed mostly adult or terrestrial insects and immature stages of aquatic insects. Included among the insect taxa were Odonata, Ephemeroptera, Plecoptera, Orthoptera, Hemiptera, Trichoptera, Lepidoptera, Coleoptera, Diptera (including midges), and Hymenoptera. More recently, Schaffter et al.² (1983) found that juvenile chinook salmon captured from the river between Red Bluff and Chico fed primarily on midges, baetid mayflies, and Aphididae. In August 1990, identifiable gut contents of 29 juvenile chinook salmon from Lake Redding on the upper Sacramento River consisted mostly of Cladocera and midge larvae (M.K. Saiki, unpublished data). Juvenile chinook salmon from the lower American River consumed all life stages of midges, nymphs, and adults of baetid mayflies and larvae of hydropsychid caddisflies (Merz and Vanicek

1996). Finally, Moore³ (1997) reported that juvenile salmon in the Sacramento River at Red Bluff fed mostly on midges, baetid and ephemereid mayflies, hydropsychid caddisflies, and perlodid stoneflies, whereas juvenile salmon in Dibble and Blue Tent creeks fed mostly on midges, baetid mayflies, capniid stoneflies, and larval fish. These observations are generally consistent with findings from our study, which showed that gut contents of juvenile salmon were composed mostly of immature aquatic insects – especially midge larvae and pupae and hydropsychid caddisfly larvae – and amorphous organic matter.

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